**CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT**

Course Code and Title**: S-CSIS414 – MACHINE LEARNING 2 LABORATORY**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

**DIRECTIONS:** Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

| Format, parts, and description | Compliance | |
|---|---|---|
| | **Complied With** | **Not Complied With** |
| **Format** | | |
| 1. The 2023-2024 syllabus format is used. | / | |
| 2. The font used in the syllabus preparation is Times New Roman (11). | / | |
| 3. DLSUD (being first) and college logos are beside each other at the top (center). | / | |
| 4. Preliminaries include all the contact details of the professor, such as the day and time. | / | |
| 5. Each WEEK has a corresponding date. | / | |
| **Content** | | |
| 6. The Learning Outcomes have more HOTS than LOTS. | / | |
| 7. CLOs and TLOs are identified for each module/lesson/topic. | / | |
| 8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities. | / | |
| 9. The journal articles/references are within the last five years. | / | |
| 10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies. | / | |
| 11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students. | / | |
| **Assessments** | | |
| 12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus. | / | |
| 13. Formative Assessments are not reflected anymore in the syllabus. | / | |
| 14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities. | / | |
| 15. The syllabus presents one-module(lesson)-one-assessment correspondence. | / | |
| 16. Only one Enabling Assessment is reflected in a given week | / | |
| 17. The Enabling Assessments are aligned with the identified CLOs and TLOs. | / | |

| | | |
|---|---|---|
| 18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term. | / | |
| 19. Thirty percent (30%) is allotted for Summative Assessments per term. | / | |
| 20. Twenty percent (20%) is allotted for Class Participation/Attendance per term. | / | |
| 21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade. | / | |
| 22. Summative Assessments are aligned with the identified CLOs, TLOs and teachinglearning activities. | / | |
| 23. Enabling Assessments are aligned with the identified CLOs and TLOs and teachinglearning activities. | / | |
| 24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD | / | |
| 25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD. | / | |
| **References and Curriculum Map** | | |
| 26. The References must be within the last 5 years except for canons and seminal studies. | / | |
| 27. It must be a combination of books, eBooks, journal articles, videos, etc. | / | |
| 28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS. | / | |
| 29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs) | / | |

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

**Self-reviewed:**

***Rolando B. Barrameda***

**Checked, verified, and endorsed:**

***Rolando B. Barrameda***
Dept. Curriculum Committee Reviewer

**Checked, verified, and approved:**

***Josephine T. Eduardo***
Department Chair

***Dr. Rubie M. Causaren***
College Dean



**De La Salle University - Dasmariñas**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER STUDIES DEPARTMENT**

# COURSE SYLLABUS

| | | |
|---|---|---|
| **COURSE CODE** | **:** | S-CSIS414 |
| **COURSE TITLE** | **:** | Machine Learning 2 |
| **COURSE TYPE** | **:** | Laboratory |
| **LEARNING MODALITY:** | | Traditional Blended Learning |
| **COURSE CREDIT** | **:** | 1 unit |
| **PRE-REQUISITES** | **:** | S-CSIS311 |
| **CO-REQUISITES** | **:** | None |
| **INSTRUCTOR** | **:** | Flordeliza Fernandez |
| **CONSULTATION** | **:** | 1:00 – 5:00 pm Friday |
| | | ffernandez@dlsud.edu.ph |
| | | (046)4811900 local 3134 |

## COURSE DESCRIPTION:

This course deals with advanced study on how to build computer systems that learn from experience. It is a subfield of Artificial Intelligence and intersects with statistics, cognitive science, information theory, and probability theory, among others. The course explains on how to build systems that learn and adapt using real-world applications from industry and scientific study.

## COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. understand the intuition behind neural network;
CLO2. demonstrate how to assess data-driven models and how to visualize the outcomes of algorithms;
CLO3. build a powerful Machine Learning models and understand how to combine them to resolve the problem;
CLO4. apply the machine learning classification and clustering algorithms; and

CLO5. formulate optimal models and use TensorFlow coding methods to address real-world issues;

**LEARNING PLAN:**

**Learning Outcomes for Midterm Period**

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|
| CLO1. Understand the intuition behind neural network | • TLO1. Explain the artificial intelligence and machine learning.<br>• TLO2. Distinguish artificial intelligence learning methods.<br>• TLO3. Demonstrate and apply data<br>• preprocessing tools. |
| CLO2. Demonstrate how to evaluate data driven models. | • TLO4. Apply working with datasets, splitting and feature calling.<br>• TLO5. Determine and generate the corpus data models |
| CLO3. Build a powerful Machine Learning models and understand how to combine them to resolve the problem | • TL06. Support observation to best fitting the data<br>• TLO7. Enhance the effectiveness of algorithm by transforming them while adding intelligent features. |
| CL04. Apply the machine learning classification and clustering algorithms. | • TLO8. Exhibits and demonstrate data observation.<br>• TLO9. Generate and evaluate data observation. |
| CLO5. Formulate optimal models and use TensorFlow coding methods to address real-world issues. | • TLO10. Collaboratively develop application and create a research paper project. |

| Module No and Learning Outcome(s) | Week | Teaching-Learning Activities / Assessment Strategies<br>Online Module Activities | Technology Enabler | Onsite | Summative | Allotted Hours |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| Module 1<br><br>TLO 1<br>TLO 2<br>TLO 3<br>TLO 4<br>CLO 1<br>CLO 2<br>CLO 5 | 1 – 2<br>8/14-8/26 | • Gospel reading and reflection<br>• Syllabus uploaded<br>• Presentation of course introduction<br>• Course requirement details | Schoolbook<br>PowerPoint | / | | 2.0 |
| | | • Gospel reading and reflection<br>• Warm-up<br>• Deep Learning Overview<br>Application of Deep Learning in different domain<br>Algorithm behind DL | Schoolbook<br>MS Teams | | | 2.0 |
| Module 2<br><br>TLO 3<br>TLO 4<br>CLO 2<br>CLO 5 | 3<br>8/28-9/2 | ✓ Gospel reading<br>Lesson Recap<br>✓ **Neural Network**<br>✓ Hopfield Neural Network<br>✓ Neural Network with Backpropagation Theory<br>Single Perception Model | Schoolbook<br>PowerPoint | / | | 2.0 |
| Module 3<br><br>TLO 5<br>TLO 6<br>CLO 3<br>CLO 5 | 4<br>9/4-9/9 | ✓ Gospel reading<br>• Lesson Recap<br>✓ **Regression and Classification of Model**<br>✓ Evaluating Regression Model Performance<br>• | Schoolbook | | | 2.0 |
| | 5<br>9/11-9/16 | | Schoolbook<br>PowerPoint | / | | 2.0 |
| | 6<br>9/18-9/23 | | | | | 2.0 |
| | 7<br>9/25-9/30 | Gospel reading<br>Lesson Recap<br>**Re-enforcement Learning**<br>Upper Confidence Bound<br>**Enabling Assessment:**<br>• Drills and Demonstration on UCP | Schoolbook<br>PowerPoint | / | | 2.0 |
| | 8<br>10/2-10/7 | • Gospel reading and reflection<br>• **Dimensionality Reduction**<br>• Principal Component Analysis (PCA) | Schoolbook | | | 2.0 |
| | 9 10/9-10/14 | • SUMMATIVE ASSESSMENT (Module 1-4) | Schoolbook | / | / | 2.0 |
| | | | ***TOTAL*** | ***16.0*** | ***2.0*** | ***18.0*** |

**Learning Outcomes for Final Period:**

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|

| | |
|---|---|
| CLO3. Develop an appreciation for what is involved in learning models from data. | • TL06. Support observation to best fitting the data<br>• TLO7. Classify domain areas of machine learning |
| CL04. Apply the algorithm to a real-world problem and other domain areas. | • TL08. Exhibits and demonstrate data observation.<br>• TLO9. Generate and evaluate data observation using python. |

| Module No and Learning Outcome(s) | Week | Teaching-Learning Activities / Assessment Strategies | Technology Enabler | Onsite | Summative | Allotted Hours |
|---|---|---|---|---|---|---|
| | | **Online Module Activities** | | | | |
| Module 4<br>TLO 5<br>TLO 6<br>CLO 3<br>CLO 5 | 10<br>10/16-10/21 | Linear Discriminant Analysis (LDA) | Schoolbook | / | | 2.0 |
| | 11<br>10/23-10/28 | ✓ Deep Learning<br>Application of real-world problem and model solution | | / | | 2.0 |
| Module 5<br>TLO 5<br>TLO 6<br>CLO 3<br>CLO 5 | 12<br>10/30-11/4 | • Gospel reading<br>• Lesson Recap<br>• TensorFlow Mobile Application and Image Classifier | Schoolbook | / | | 2.0 |
| | 13-14<br>11/6-11/18 | • Gospel reading and reflection<br>• Research Collaboration<br>• Enabling assessment: Chapter 1,2 and 3 | Schoolbook<br>PowerPoint | / | | 4.0 |
| Module 6<br>TLO 5<br>TLO 6<br>CLO 3<br>CLO 5 | 15<br>11/20-11/25 | Gospel reading and reflection<br>Research Collaboration<br>Enabling assessment: Chapter 4,5,6 and 7 | Schoolbook | / | | 2.0 |
| | 16<br>11/27-12/2 | • Forum, Collaboration, Research Presentation<br>• Enabling assessment: Paper Submission Conference (Local / International) | Schoolbook<br>PowerPoint | / | | 2.0 |
| | 17<br>12/4-12/9 | • Completion of Final Output | Schoolbook | / | | 2.0 |

| | | | | | | |
|---|---|---|---|---|---|---|
| | 18<br>12/11-<br>12/16 | • SUMMATIVE ASSESSMENT (Module 5-7) | Schoolbook | | / | 2.0 |
| | | | ***TOTAL*** | ***16.0*** | ***2.0*** | ***18.0*** |

**GRADING SYSTEM:**

**Midterm**

| | |
|---|---|
| Attendance/Class Participation | - 20% |
| Enabling Assessments | - 50% |
| Summative Assessments | - 30% |
| | 100% |

**Finals**

| | |
|---|---|
| Attendance/Class Participation | - 20% |
| Enabling Assessments | - 50% |
| Summative Assessments | - 30% |
| | 100% |

**Overall Final Grade =** $\frac{\text{Midterm + Final}}{2}$

**COURSE POLICIES AND REQUIREMENTS:**

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes are delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.

iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be always honest; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

## REFERENCES:

### On-Site References


| **DLSU-D E- Books References:** | |
|---|---|
| Accession Number: 2363562 | Campesato, Oswald. 2020. Artificial Intelligence, Machine Learning, and Deep Learning. Mercury Learning & Information. |
| Accession Number: 2145644 | Taylor Smith. 2019. Supervised Machine Learning with Python : Develop Rich Python Coding Practices While Exploring Supervised Machine Learning. Birmingham : Packt Publishing. |

| | |
|---|---|
| Accession Number: 1703794 | Jianing Wei. , Anurag Bhardwaj., Wei Di . 2018. Deep Learning Essentials : Your Hands-on Guide to the Fundamentals of Deep Learning and Neural Network Modeling. Birmingham, UK : Packt Publishing |
| **Publication Paper:** | |
| DOI: 10.1109/ICMLA52953.2021.00197 | Agnes Tegen; Paul Davidsson; Jan A. Persson.2021. Active Learning and Machine Teaching for Online Learning: A Study of Attention and Labelling Cost. Published in: 2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA) |
| DOI: 10.1109/MLSP55214.2022.9943373 | Goutam Kumar Sahoo; Keerthana Kanike; Santos Kumar Das; Poonam Singh. 2022. Machine Learning-Based Heart Disease Prediction: A Study for Home Personalized Care. Published in: 2022 IEEE 32nd International Workshop on Machine Learning for Signal Processing (MLSP) |

**Prepared:** **Computer Science Department**
1st Semester 2023-2024

**Approved:** **Josephine T. Eduardo**
Chair, Computer Studies Department

**Endorsed:** **Rubie M. Causaren, PhD**
Dean, College of Science and Computer Studies

**De La Salle University - Dasmariñas**

**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER SCIENCE DEPARTMENT**

**PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE**

**COURSE CODE** : S-CSIS414
**COURSE TITLE** : MACHINE LEARNING 2
**COURSE TYPE** : Lecture
**COURSE CREDIT** : 2 Units

| Course Learning Outcome | Program Learning Outcome | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | PLO 1 | PLO 2 | PLO 3 | PLO 4 | PLO 5 | PLO 6 | PLO 7 | PLO 8 | PLO 9 |
| 1. Understand the intuition behind neural network | / | / | / | | / | | | | |
| 2. Demonstrate how to assess data-driven models and how to visualize the outcomes of algorithms. | | / | / | / | / | | / | | |
| 3. Build a powerful Machine Learning models and understand how to combine them to resolve the problem. | | / | / | | / | | / | / | |
| 4. Apply the machine learning classification and clustering algorithms. | / | / | / | / | / | / | / | / | / |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 5. Formulate optimal models and use TensorFlow coding methods to address real-world issues. | | | | / | / | | / | / | / |

*NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)*

**De La Salle University - Dasmariñas**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES COMPUTER SCIENCE DEPARTMENT**

## COMMON GE LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

**COURSE CODE** : S-CSIS3414

**COURSE TITLE** : MACHINE LEARNING 2

**COURSE TYPE** : Lecture

**COURSE CREDIT** : 2 Units

| Course Learning Outcome | Common GE Learning Outcome | | | | |
|---|---|---|---|---|---|
| | CGE1 | CGE2 | CGE3 | CGE4 | CGE5 |
| 1. Understand the intuition behind neural network | / | | | / | / |
| 2. Demonstrate how to assess data-driven models and how to visualize the outcomes of algorithms. | / | | | / | / |
| 3. Build a powerful Machine Learning models and understand how to combine them to resolve the problem. | | | | / | / |
| 4. Apply the machine learning classification and clustering algorithms. | | | / | / | / |
| 5. Formulate optimal models and use TensorFlow coding methods to address real- world issues. | / | | / | / | / |

*NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)*

**COMMON GE LEARNING OUTCOME**

1. Critiques/evaluates individual's character, actions, and decisions based on Christian-like perspective
2. Assesses situations and proposes ways that benefit the country, particularly its environment and its marginalized sector
3. Explains in various occasions the relationship of the course/subject with the other subjects within or outside the discipline
4. Recommends ways to improve the subject/course

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## GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

| Program Learning Outcome | Graduate Attributes/Institutional Learning Outcomes | | | | |
|---|---|---|---|---|---|
| | GA1 | GA2 | GA3 | GA4 | GA5 |
| 1. Apply concepts of computing and mathematics in different domains of computer science. | | | | / | / |
| 2. Identify and analyze problems and define computing requirements appropriate to their solutions. | | | | | / |
| 3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources. | | | / | / | / |
| 4. Document computer-based systems by writing effective reports for risk- assessment impact. | | | | / | / |
| 5. Work effectively as an individual or as a member of a team during software development. | | | / | / | / |
| 6. Communicate effectively with the computing community and non-computer science professionals about computing activities. | | | | / | / |
| 7. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues. | / | / | / | | |
| 8. Engage in continuing professional development through performing independent or collaborative research as computing professionals. | | | / | / | / |

*NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.*

| Graduate Attribute (GA) | De criptors (Institutional Learning Outcome)[1] |
|---|---|
| 1. God-centered | a) Creates an environment where the experience of God is lived and shared<br>b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives)<br>c) Observes and maintains ethical standards in dealing with the different stakeholders<br>d) Integrates Christian perspectives and values in all undertakings<br>e) Manifests humility and respect in relating with other people |
| 2. Patriotic | a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives<br>b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts<br>c) Engages actively in political, social, economic, and cultural transformation for nation building<br>d) Brings pride and honor to the community and the country<br>e) Patronizes locally produced products and promotes them globally |
| 3. Socially Responsive | a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation<br>b) Engages in ecological advocacies as responsible stewards of God's creation<br>c) Continuously works in solidarity with people and institutions to effect liberating action<br>d) Brings forth awareness and deeper understanding of social realities<br>e) Manifests spirit of volunteerism through social actions that improve communities |
| 4. Globally Competitive | a) Takes progressive responsibility for own learning and development<br>b) Promotes critical and creative thinking, self-knowledge and self-mastery<br>c) Is committed to rise above mediocrity<br>d) Shows dedication, commitment and love for work to attain exceptional results<br>e) Exhibits innovativeness and creativity in various contexts |
| 5. Professionally Competent | a) Possesses leadership qualities of **C**ompetence, **H**onesty, **A**ccountability, **R**esponsibility and **T**ransparency (CHART)<br>b) Effectively communicates in various forms of media<br>c) Strives to be at the top of their chosen fields<br>d) Effectively works and collaborates in a multi-disciplinary team<br>e) Adapts to changes in the workplace, local society and global community |